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Research paper

Transmission line redundancy for grid resilience enhancement: The concept of Transmission Lines contributing to Energy Not Supplied (TLENS) on Malawi's transmission grid

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ABSTRACT

Enhancing power grid resilience, such as redundancy, is crucial for ensuring a reliable energy supply amid increasing environmental challenges. Transmission line redundancy involves identifying lines to be prioritised for improvement and how to improve the identified lines. The challenge remains to identify critical lines using specified transmission level indicators that quantify how the failure of lines affects electricity delivery and factors to consider when planning for redundancy. This paper introduces a novel resilience indicator called Transmission Lines' Contribution to Energy Not Supplied (TLENS), which measures how specific transmission line failures impact overall energy not supplied (ENS). Using TLENS, critical lines were identified, and redundancy effectiveness was evaluated. Further, TLENS was integrated with network topology, disaster exposure maps and stakeholder consultations to formulate and prioritise redundancy choices. Using DIgSILENT PowerFactory, a case study was conducted to observe grid behaviour through its reliability analysis toolbox. The study identified Nanjoka – Nkhotakota, Nkhotakota-Dwangwa, Dwangwa-Chintheche and Kapichila – Mlambe as substantial ENS contributors, posing significant resilience risks. Combining multiple redundancy scenarios achieved the lowest TLENS. However, regions prone to disasters might benefit more from either loop connections (alternative supply routes) than parallel line redundancy and distributed generation or making the lines more robust. This research offers a comprehensive framework for future grid resilience efforts and recommends exploring the comprehensive cost-benefit analysis of viable redundancies.

1. Introduction

The resilience of electrical grids has achieved rising importance due to the growing demand for reliable electricity and the rising frequency of extreme weather events and other disruptions. The basic resilience perspective is segregated into three portions — hard, soft, and mixed (Hossain et al., 2021). Hard resilience describes the physical aspects of functionality, such as engineering and ecological systems. Soft resilience is the humane and social network, which comprises culture, governance, and social models, which are fundamental to understanding the effects of disasters and the practicality of emergency management systems. Mixed resilience pools different infrastructures to understand the structures, from hard to soft resilience. Grid resilience can further be classified based on its performance, implementation, temporality, interdependency, and occurrences of events (Hossain et al., 2021). Resilience definitions demonstrate that resilience is a function of time, and, as such, it can be split into *short-* and *long-term* resilience. Short-term methods represent the preventive and corrective techniques employed before, during, and after extreme weather events. Measures that can be taken involve reserve planning, generation dispatch, guaranteeing black-start capacities, coordination with neighbouring grids, energy storage, and demand-side management. The *long-term* techniques describe the long-term adaptability planning for improving the network's resilience to impending severe events and climate change. Redundant transmission lines are under the umbrella of long-term measures (Panteli and Mancarella, 2017).

The research on improving the resilience of power systems can be divided into three stages: prevention before, resistance and absorption during disruptions, and recovery following disruptions. Prevention strategies can be divided into two main categories: operation and

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Table 1
List of references and respective transmission line hardening measures.

Reference	Transmission line hardening measures						
	Overhead line undergrounding	Transmission line upgrade	Redundancy	Vegetation management	Tie lines	DERs	Line rerouting
(Yang and Li, 2022)			✓	✓			
(Xu et al., 2023)	✓	✓	✓				
(Pan et al., 2022)			✓				
(Shi et al., 2018)			✓				
(Ding et al., 2022)					✓	✓	
(Panteli et al., 2017; Ding et al., 2022)		✓					
(Najafi Tari et al., 2021; Wang et al., 2016)		✓					
(Wang et al., 2016; Taylor et al., 2022)				✓			
(Najafi Tari et al., 2021; Wang et al., 2016; Group and Bresch, 2010)							✓
(Hossain et al., 2021; Feng et al., 2022; Najafi Tari et al., 2021)	✓						

planning measures. System operators can enhance the survivability of power systems by generation rescheduling, topology variation, and demand-side management before or during severe disruptions. Planners can expand equipment redundancy or improve the disaster survival level of equipment to withstand disruptions (Zhang et al., 2024). Strategies to improve resilience can also be divided based on the timeline of the disruptive event, such as preventive measures implemented before the event's start, corrective measures executed during the event's progression, and restorative measures adopted after the event. Grid-hardening measures are usually employed long before the onset of an event. Existing transmission line enhancement measures include overhead line undergrounding (Hossain et al., 2021; Xu et al., 2023; Feng et al., 2022; Najafi Tari et al., 2021), design standards upgrade (Xu et al., 2023; Najafi Tari et al., 2021; Wang et al., 2016; Panteli et al., 2017), line rerouting (Najafi Tari et al., 2021; Wang et al., 2016; Group and Bresch, 2010), tree pruning or vegetative management (Wang et al., 2016; Yang and Li, 2022; Taylor et al., 2022), tie lines (Ding et al., 2022), DERs (Ding et al., 2022), and redundancy (Panteli et al., 2017; Yang and Li, 2022; Yang et al., 2018; Pan et al., 2022; Shi et al., 2018). Overhead line undergrounding can be selective or full (Hossain et al., 2021; Najafi Tari et al., 2021) or strategic (Feng et al., 2022). Upgrading the components with more robust materials constitutes a primary hardening strategy to make the features more robust to severe weather (Panteli and Mancarella, 2015a). However, disturbances may exert additional load that may exceed the design strength of the material and cause component breakdown (Lu et al., 2018). Line rerouting is the most effective approach for reducing the exposure of components to disasters, but the cost of rerouting may be prohibitive (Aziz et al., 2021). Tree cutting is effective in areas exposed to wind disasters as these strong winds can bring down trees, which could fall on OHL (Panteli and Mancarella, 2015b). Redundancy, which is one of the principles of resilience theories (Shi et al., 2018), is a long-term resilience measure. Redundancy ensures functional continuity, increases the network's reliability and improves the network's adaptive capacity and resilience (Shi et al., 2018). Redundancy also helps to lengthen the maintenance cycle and lower average costs (Pan et al., 2022). Table 1 Summarises studies in transmission line hardening measures.

(Daeli and Mohagheghi, 2022) This work focuses on planning measures for improving power system resilience through preventive transmission line redundancy. Transmission line redundancy, which includes additional transmission paths or lines that provide alternative routes for power delivery in the event of a primary path failure, becomes a pivotal strategy in enhancing grid resilience (Yang and Li, 2022; Shi et al., 2018; Daeli and Mohagheghi, 2022). Further, redundant transmission lines in areas with high failure rates can lower the failure rate of the transmission system (Yang and Li, 2022). Redundancy, sometimes referred to as resourcefulness, is one of the four key characteristics of resilience besides resistance (robustness), reliability (availability), and response and recovery (rapidity) (Panteli and Mancarella, 2017; Panteli et al., 2015a) and is an important property to ensure the stability of power grids (Plietzsch et al., 2016).

Although many studies have been conducted to enhance power system resilience from the generation, transmission, and consumption perspectives, long-term, preventive planning methods that consider identifying critical transmission lines and modelling their redundancies still need further in-depth research. Compared with existing methods, the novelty of the proposed approach lies in three aspects:

1) an introduction of a novel quantitative resilience indicator, the Transmission Lines Contributing to Energy Not Supplied (TLENS).
2) identification of critical lines using TLENS, which are candidates for transmission line redundancy.
3) demonstration of the effect of transmission line redundancy on power grid resilience.
4) integrating it with other qualitative aspects to determine the viable redundancy scenarios.

This differs from previous works (Panteli and Mancarella, 2017; Yang and Li, 2022; Panteli et al., 2015a; Plietzsch et al., 2016; Soikkeli et al., 2023; Rodrigues Mendes Ribeiro et al., 2019; Nagarajan et al., 2016; Kim et al., 2017; Zhao et al., 2021) especially in using TLENS to identify critical lines. This is based on the premise that transmission hardening involves two key considerations: 1) how to strengthen the lines, and 2) which lines should be prioritised for enhancement (Xu et al., 2023). The robustness and redundancy in networks under different network conditions using loss of load frequency (LOLF) and loss of load expectation (LOLE) were compared (Panteli and Mancarella, 2017). The non-linearity between the cost-benefit and the increase of redundancies in the system was demonstrated by comparing local and global redundancy in the cost/redundancy optimisation (Plietzsch et al., 2016). The optimal level of redundancy for production components in a system whose output performance can be affected by cyber-attacks was estimated (Soikkeli et al., 2023). The benefits of adding redundancy to the system (Rodrigues Mendes Ribeiro et al., 2019), and computationally efficient algorithms for designing resilient transmission systems (Nagarajan et al., 2016) were assessed. The lowest cost-effectiveness ratio and best resilience improvement effect were used to select the best transmission line redundant strategies (Yang and Li, 2022). Finally, the decrease in the probability of failure was used to evaluate the effectiveness of redundant transmission lines (Zhao et al., 2021).

Although the above analysis considers transmission line redundancy, redundancy identification using TLENS and demonstrating the effectiveness of redundant transmission lines using TLENS, to our knowledge, has not been presented in the literature. The approach used to identify critical lines was not mentioned in (Panteli and Mancarella, 2017; Panteli et al., 2015a). Frequency stability (cascading failures) was used

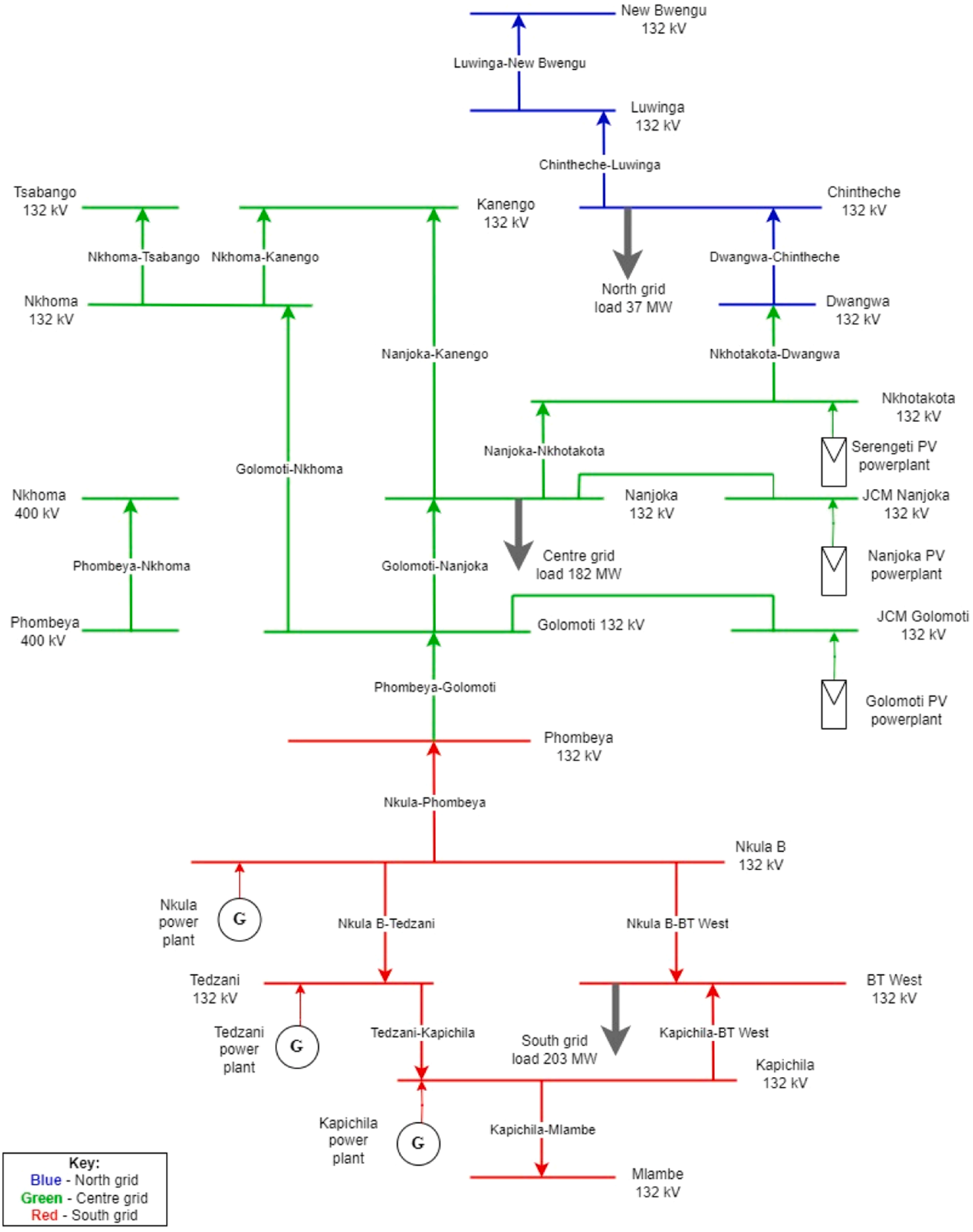


Fig. 1. Malawi's transmission network model used in this paper.

to determine the need for global (local) redundancy (Plietzsch et al., 2016). Time until the loss was employed to evaluate the cost effectiveness of redundant locations, highlighting the event frequency required to make a redundant allocation financially viable (Soikkeli et al., 2023). An overloaded transmission system was used to indicate the need for redundancy, and a selection was made from a set of potential upgrades (Nagarajan et al., 2016). The availability of redundant lines was assessed in (Kim et al., 2017). The larger value of the transmission-level resilience

indicator reflected that the faulty transmission line caused severe damage to the system and was essential to strengthen it for the overall recovery of the system (Yang and Li, 2022). However, the transmission level indicator was generalised. In (Zhao et al., 2021), a higher probability of transmission line failure or the larger load shedding due to this transmission line indicated the criticality of this line and thus needed to be enhanced. However, the impact of line failure on customers was not quantified. Identifying these lines helps utility companies prioritise where to invest in upgrades and maintenance to improve grid resilience. Moreover, understanding which lines have the greatest impact on ENS can help in planning redundancy and network design to minimise the impact of outages. This enhances the optimal and effective resource allocation to reduce the frequency and duration of outages. This knowledge is crucial in developing effective emergency response and recovery plans. The grid can maintain service continuity, minimise disruption, and expedite recovery by incorporating redundancy, although the degree of redundancy and its effectiveness must be carefully planned and managed. An essential indicator in this planning is the concept of the novel TLENS. It evaluates the extent to which the failure of a particular transmission line contributes to energy not supplied (ENS), providing a clear understanding of the criticality of each transmission line. By understanding how much energy might not be delivered to customers due to the failure of a transmission line *x,* decision-makers can invest in areas that would improve the line's robustness. Further, TLENS can guide decisions that prioritise redundancy or diversification to reduce the impact of line failure as opposed to economically efficient strategies. Furthermore, TLENS directly quantifies how much energy customers lose during disturbances, which can lead to possible financial losses, reduced productivity, and societal harm. Consequently, decisions shift towards strengthening identified critical lines, leading to a targeted infrastructure investment. This indicator, thus, allows for a targeted approach to improving redundancy, highlighting lines that, if reinforced or replicated, could significantly enhance grid resilience. A resilient grid maintains its functions even during disruptions and is restored as quickly as possible when disrupted (Xu et al., 2023). Thus, TLENS demonstrates an interaction of absorptive, restorative and adaptive resilience features. The higher the TLENS, the higher the reduction in system functionality and the longer the restoration time.

This paper demonstrates the effectiveness of intentional transmission line redundancy in strengthening grid resilience using the TLENS metric. The primary objectives were as follows: i) Assess the vulnerability of transmission lines by analysing the existing transmission network to identify lines with high TLENS values, indicating a critical role in grid reliability; ii) Develop and evaluate redundancy strategies tailored to lines with high TLENS values to minimise the impact of line failures on energy supply; and iii) Apply the proposed redundancy strategies and TLENS metric in a practical case study to demonstrate the effectiveness of the approach.

The paper has been structured as follows: Section 2 details the methodology for identifying TLENS, proposing potential redundant routes based on topology and accessibility to existing road networks and evaluating the effectiveness of proposed redundancies. Section 3 presents a criticality assessment to show TLENS and redundancy strategies and discusses the effectiveness of redundancy through a practical case study. Section 4 provides conclusions. By addressing transmission line redundancy through the lens of TLENS, this paper intends to contribute to developing robust, resilient power systems capable of withstanding and recovering from disruptions efficiently.

2. Methodology

A comprehensive and systematic approach is necessary to improve the electrical network's resilience through improved transmission line redundancy. This methodology outlines the step-by-step process undertaken to evaluate the present infrastructure, identify weaknesses, design and implement redundancy measures, and confirm measures.

Table 2
Generation plants and their installed capacities and locations.

Generation plants	Installed capacity (MW)	Location (transmission grid)
Nkula hydro	135	South
Tedzani hydro	121	South
Kapichila hydro	129.6	South
Wovwe hydro	4.5	North
Ruo – Ndiza hydro	8.2	South
Golomoti solar	20	Centre
Nanjoka solar	60	Centre
Serengeti solar	20	Centre
Mapanga diesel	20	South
Luwinga diesel	10	North
Kanengo diesel	6	Centre
Chichiri diesel	36	South

The proposed methodology aims to create a more resilient grid capable of withstanding and quickly recovering from various potential disruptions. The following sections detail the specific steps involved in this process, from initial data collection and risk analysis to the implementation and ongoing refinement of redundancy strategies.

2.1. Power system model

This scientific paper used a detailed model of Malawi's power system, as depicted in Fig. 1. As of November 2022, Malawi's power system was composed of 172 buses in five voltage levels: 400 kV, 132 kV, 66 kV, 33 kV, 11 kV and 0.4 kV. It comprises the generation, transmission, distribution, and load centres. The generation plants' capacity and locations are presented in Table 2. System improvements could have occurred since the date this data was collected, but they were not included in this analysis.

Malawi has fifty-five load centres: fourteen in the North, twenty-two in the Centre, eighteen in the South and one in the Generation section. Capacitor banks at Nanjoka, Kanengo, Chintheche, and Karonga were switched on for this analysis to maintain grid compliance with voltage levels. The grid operator and the electricity generating company provided the model parameters. For illustration, only two transmission voltage levels have been presented in the power system model in Figure 1Furthermore, the regional loads were aggregated into three, one in each area, to represent all regional loads. The generators represented all individual generators in each power station.

Model validation was three-way. Firstly, by comparing calculation parameters against measured parameters. The average actual ENS for the grid system for the four years from January 2018 to December 2021 was 8073 MWh/a (ESCOM, 2022a). This was a close estimation of the model ENS (8392.863 MWh/a), although with an approximately 4 % error. The error could be from calculations. The transformer stochastic model could cause the calculation error, as two-year failure data was used because of data availability issues. Secondly, by the grid operator, who are custodians of the network. Thirdly, field and software expert validation was used. DIgSILENT PowerFactory (PF) software was used to run all the analyses. DIgSILENT and PowerFactory are sometimes used interchangeably.

2.2. DIgSILENT PowerFactory

DIgSILENT" is an abbreviation for "DIgital SImuLation of Electrical NeTworks" and is a computer-supported engineering tool for evaluating transmission, distribution, and industrial electrical power systems. The precision and authenticity of results attained with PowerFactory have been verified in numerous implementations by organisations involved in the planning and operation of power systems world-wide (DIgSILENT GmbH, 2023). PowerFactory includes a wide-ranging list of simulation functions, such as load flow analysis, short-circuit analysis, sensitivities, contingency analysis, quasi-dynamic simulation, network reduction, protection analysis, arc-flash hazard analysis, cable analysis, power

quality and harmonics analysis, connection request assessment, transmission network tools, distribution network tools, outage management, probabilistic analysis, reliability analysis, optimal power flow, unit commitment and dispatch optimisation, economic analysis tools, state estimation, time-domain simulation, modal / Eigenvalue Analysis, and system parameter identification. These analyses use the Monte Carlo or Quasi-Monte Carlo-based approach (DIgSILENT GmbH, 2023).

Reliability analysis is an automation and probabilistic expansion of contingency analysis. The reliability evaluation process considers the network topology, protection systems, constraints, stochastic failure and repair models to produce reliability indicators, including TLENS. The generation of reliability indices consists of creating stochastic models, defining feeders, configuring switches, and defining loads in that order. Optionally, load characteristics, protection devices, maintenance plans, and constraints may be defined, created, or specified.

PowerFactory is principally intended to be used and operated in a graphical environment. Data is entered by drawing the network elements and then editing and assigning data to these objects. In addition to the basic data, other required data depend on the type of analysis. The reliability analysis toolbox in DIgSILENT, from where TLENS are extracted, requires system reliability data, such as failure and repair rates and load models (DIgSILENT GmbH, 2021).

2.3. Stochastic models: failure rates, repair rates and load models

Malawi's power system, unlike the standard models such as the RBTS (DIgSILENT GmbH, 2021), did not have network reliability data, so stochastic modelling was necessary to generate this. These models were essential for reliability analysis. Failure modelling was based on the grid operator's system performance data (ESCOM, 2022a). Standard stochastic models include transmission overhead lines (OHL), buses (terminals), transformers, generators and loads. The available data limited this study's stochastic models to OHL, transformers and loads. Because common mode failures (CMF) are also part of extreme events (Bie et al., 2017), this study's stochastic model also included CMF. The failure rates for CMF were based on system performance under extreme events (100-year cyclone event). Repair durations, outage hours, line lengths (for OHL), and number of faults were used to determine the outage expectancies and failure frequencies. Failure frequency for n similar elements is given by

$$\lambda_n = \frac{number\ of\ faults\ from n elements}{number\ of\ years * number\ of\ monitored\ equipment} \quad (1)$$

in per annum, $\frac{1}{a}$

Outage expectancy for n similar elements is given by

$$Q_n = 1 - p_n \quad (2)$$

(in hours per annum, $\frac{h}{a}$), where p_n is element n availability given by

$$p_n = \frac{Time\ To\ Failure\ for n elements(TTF_n)}{Time\ To\ Failure\ for n elements(TTF_n) + Time\ To\ Repair\ for n elements(TTR_n)} \quad (3)$$

where

$$TTF_n = \frac{1}{\lambda_n} \quad (4)$$

and

$$TTR_n = \frac{repair\ hours\ for n elements}{number\ of\ faults\ for n elements} \quad (5)$$

Table 3
Redundancy planning approaches.

Redundancy planning approach	Aim
Topological	Achieve redundancy in either parallel or loop connections Minimise exposure to other disasters Access to existing road networks
Criticality assessment (resilience calculation parameters)	Reduce line contribution to ENS
Stakeholder consultations	Most affected parts of the network (customers) Grid operator redundancy plans (expansion plans) Threats to resilience improvements (redundancy)

Load modelling determined the customers connected to each load centre (substation). The available data was based on business centres, not load centres. The calculation of customers connected to each load was based on the assumed geographical location of load centres and active power operating limits. Included in the stochastic modelling were the geographical coordinates for the load centres, transmission lines, substations and terminals. This was done to enable the visualisation of these elements in the geographical diagrams in the power system calculation tool.

2.4. Resilience assessment – assessing the criticality of transmission lines

This involved risk identification and criticality assessment. Potential risks to transmission lines, such as natural disasters and equipment failures, were evaluated. The criticality of each transmission line was assessed by analysing its role in the grid, including its contribution to ENS and its impact on overall grid stability. TLENS was used as a reliability-based infrastructure resilience indicator, the contribution being in MWh/a. Lines that contribute more to ENS can significantly affect grid resilience. Lines with high ENS values are often either more failure-prone or critical to the network and thus need to be improved. This was observed in (Zhao et al., 2021), where the more load shedding a transmission line causes, the more critical that line becomes and that these lines have a greater failure probability. On average, ENS is the total energy not delivered to the system loads.

$$ENS = \Sigma\ LPENS_i \quad (6)$$

where $LPENS$ is the Load Point Energy Not Supplied, in MWh/a, thus

$$LPENS_i = ACIT_i \cdot (Pd_i + Ps_i) \quad (7)$$

where $Pd_i$is the weighted average amount of power disconnected at the load point i, and Ps_i is the

the weighted average amount of power shed at the load point i.

$ACIT$ is the Average Customer Interruption Time in h/a, thus

$$ACIT_i = \sum_k 8760 * \Pr_k * frac_{i,k} \quad (8)$$

$\Pr_k$ is the probability of occurrence of contingency k and $frac_{i,k}$ is the fraction of the load which is lost at the load point i, for contingency k

Failure of the power system to supply critical loads during disasters is another indication of un-resilient systems (Manyena and Machingura, 2019). This failure results from compromised lines.

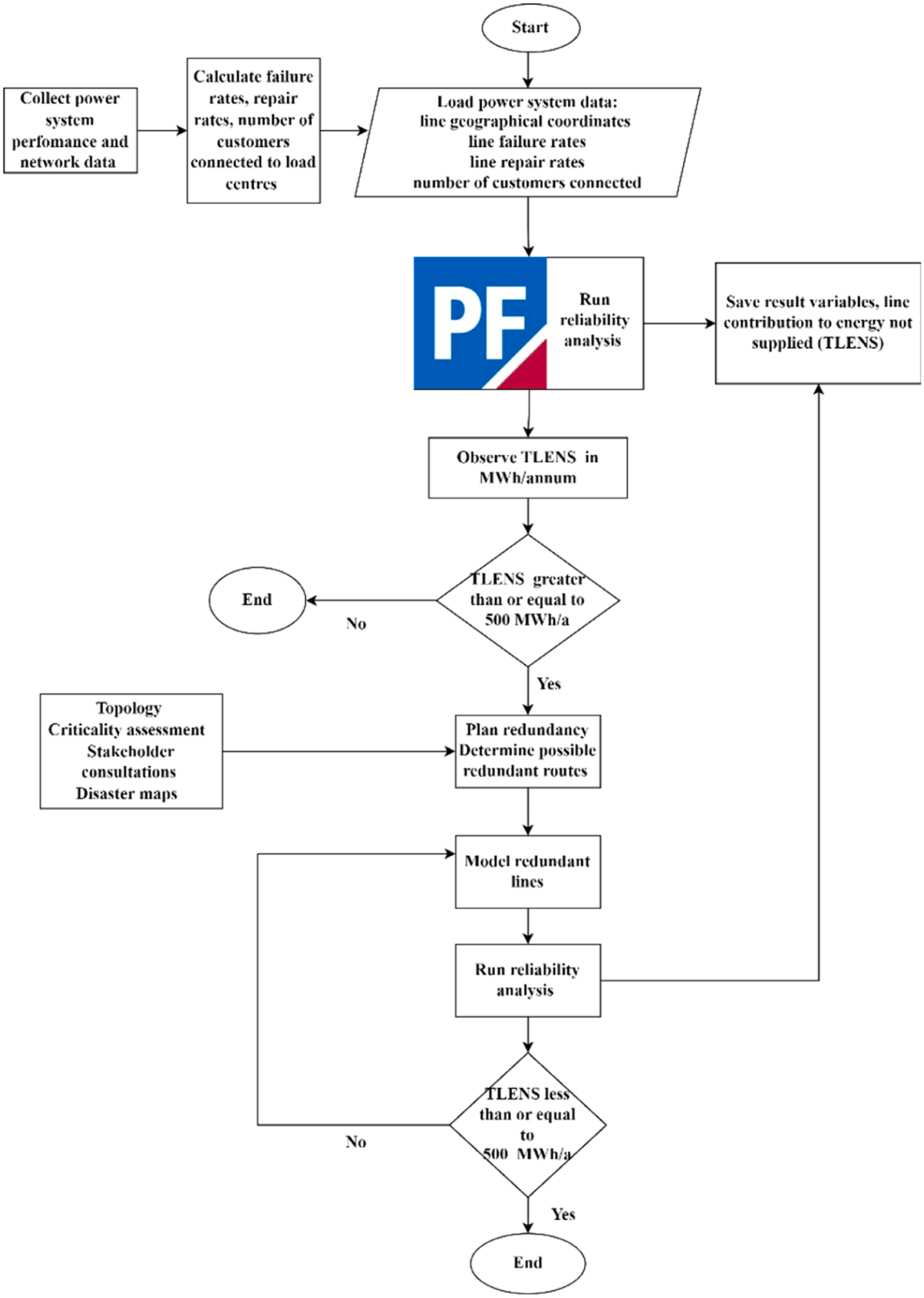
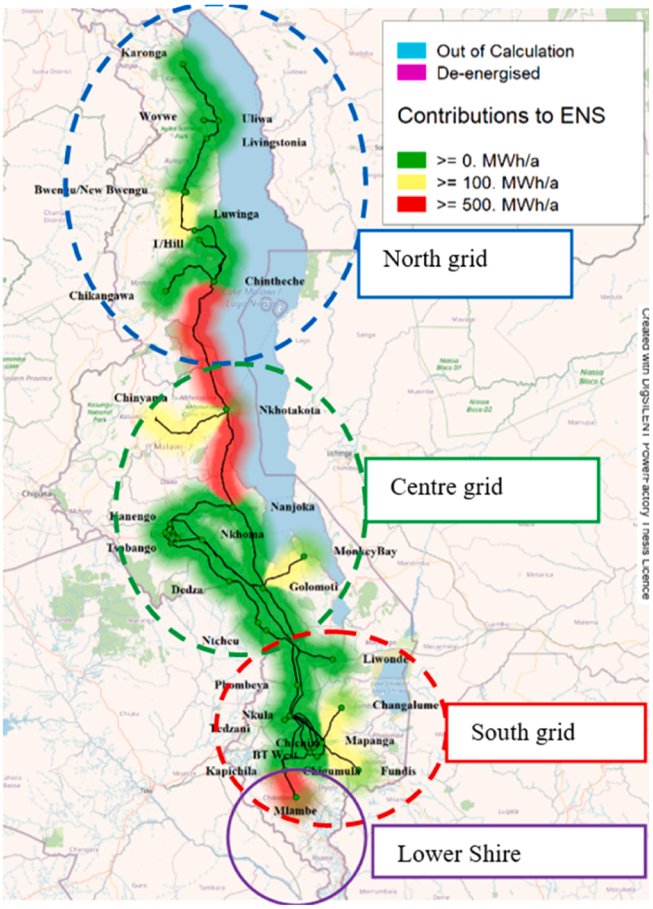


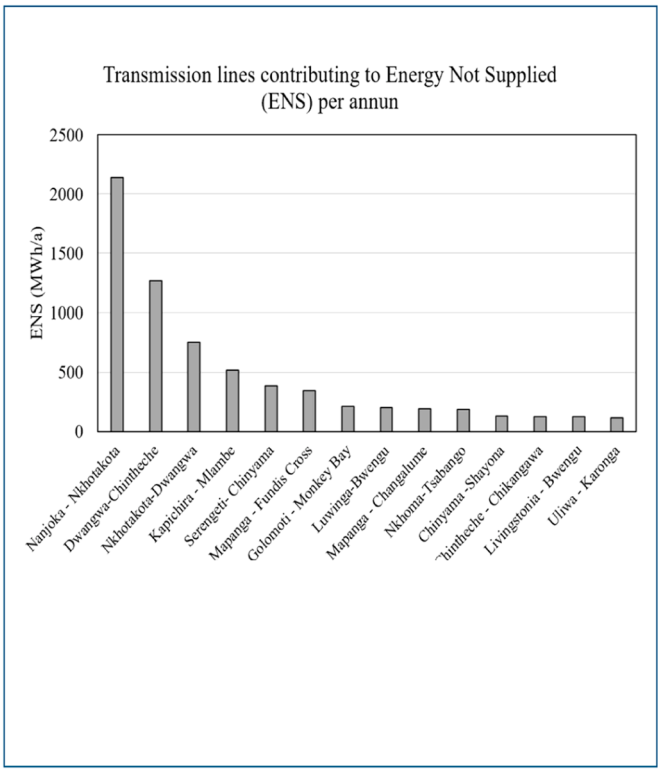
Fig. 2. Simulation process for transmission line redundancy.

$$\text{Lines contribution to ENS per annum in MWh/a} = P_{Outage_{line}} * t_{Outage_{line}} \quad (9)$$

where $P_{Outage_{line}}$ is the power not supplied during the outage of a line, and $t_{Outage_{line}}$ is the duration of the line outage in hours per annum. Actual damage data validated this analysis during the 2022 Tropical Cyclone Ana (TCA). The lines were considered resilience risks if they contributed more to the ENS. Specifically, lines that contributed more than 500 MWh/a were considered critical and thus needed enhancement. Since these transmission lines are exposed to severe events, failure of these lines can cause significant power outages to many customers and cause



a. Geographic diagram before enhancement (**BE**)



b. Typical values of energy not supplied (ENS) before enhancement (**BE**)

Fig. 3. Transmission lines contributing to ENS in MWh per annum.

widespread power outages. Actual damage data validated this analysis during TCA. The National Grid (National Grid,; Grid, 2019) targets ENS of 316 MWh/a such that the company earns money for an annual loss of supplies below the target value. To apply the methods to identify the lines contributing to ENS per annum, reliability studies were conducted in the DIgSILENT PowerFactory environment.

2.5. *Resilience enhancement planning (redundancy planning)*

Resilience improvement was focused on redundancy planning, implementation and evaluation of redundancy measures. Resilience improvement planning (redundancy) approach was based on considerations summarised in Table 3.

The topological approach involved scanning the power system network topology and using engineering judgement to ensure that the topology achieves redundancy minimises exposure to extreme events and provides easy accessibility regarding road networks. Easy access to existing road networks helps in minimising construction costs. The topological approach is in line with (Zhu et al., 2023), where alternative redundancy schemes were generated by considering network topology including location, capacity and length. In this case study, redundant lines were built on existing substations to minimise construction costs since financial constraints were major transformation setbacks (Chivunga et al., 2024). However, disaster maps were used to determine the vulnerability of redundancy options. Redundancy planning based on criticality assessment used results from line vulnerability assessment using TLENS. Lines with high TLENS were considered critical and thus needed to be enhanced. Stakeholder consultations involved conducting interviews and desk research to identify threats to redundancy plans and the most affected portions of the grid. Routes were proposed for every identified critical line and their respective destination buses following existing road networks. These routes were considered "enhancement scenarios (ES)," which included their respective destination nodes, line length in km, i-terminal, j-terminal, and type of redundancy (loop, parallel or alternate). Although all types of redundancy scenarios provide means of substituting damaged lines, some may be less favourable options, especially when disasters are inevitable. For example, parallel redundancies are exposed to the same weather conditions as opposed to loop or alternate redundancies (Panteli et al., 2015b). In addition, parallel redundancies do not offer the diversity in terms of power supply (Parallel connection and redundancy of power supplies,). The choice of redundancy scenarios depends on specific network requirements.

2.6. *Effectiveness of resilience enhancement (redundancy)*

Transmission line redundancy aimed to reduce the contribution to ENS per annum, thus decreasing the criticality and exposure of lines and consequently reducing power restoration time and magnitude of impact. After the redundancies were modelled, their effectiveness was evaluated by running reliability studies. Routes that had lower TLENS were considered viable routes. Redundancy was considered a viable route if the option reduced the ENS contribution to less than 500 MWh per annum. Multiple routes were available for some load centres, and the alternatives with the lowest contribution were considered viable routes.

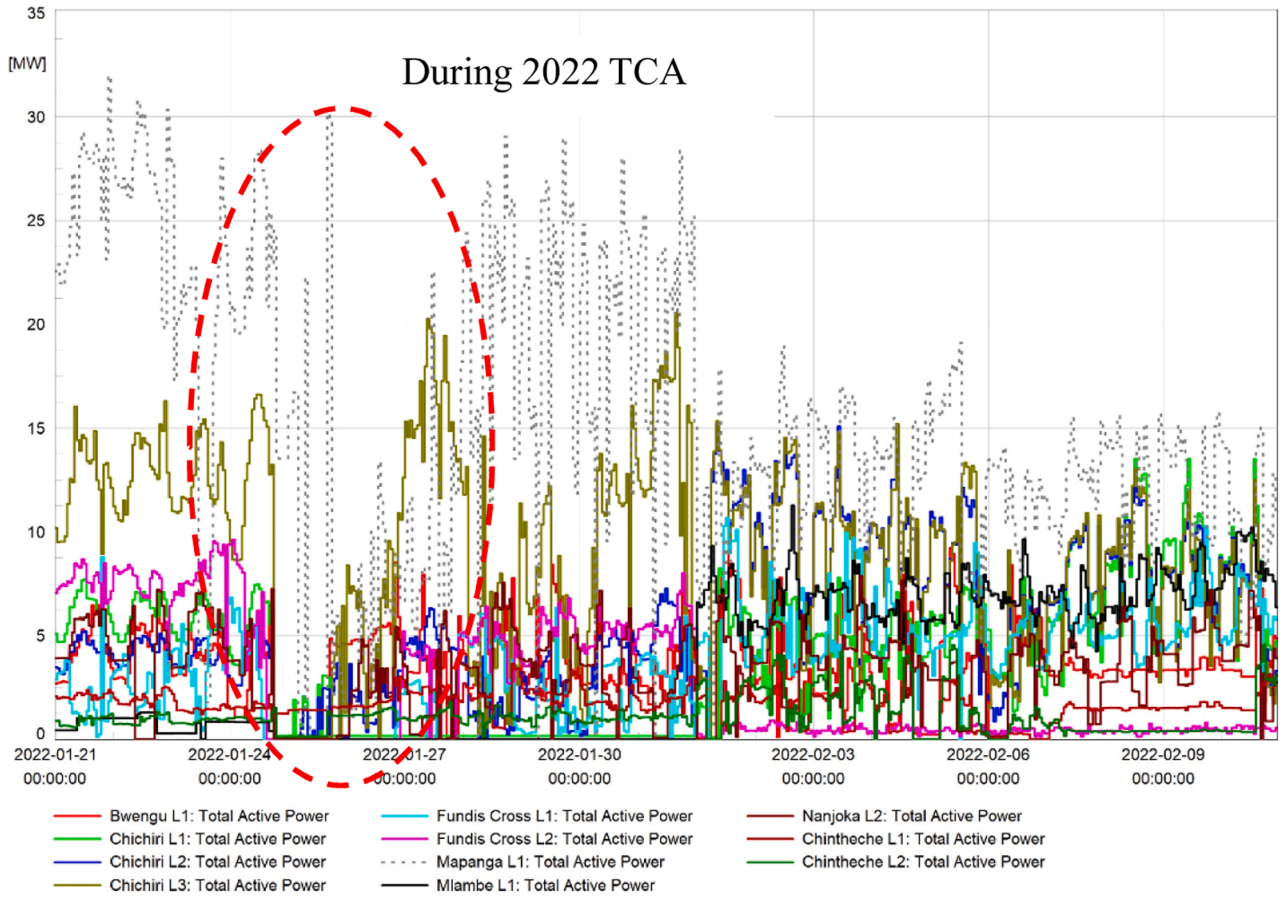


Fig. 4. Impacts of broken transmission lines on significant load centres.

Moreover, those redundancy options were considered viable if they were not exposed to other existing disasters. A combination of different ES was done to achieve a non-red thematic map. A combination of all lines that lowered the overall ENS contribution was taken on board; thus, the strategies with the lowest TLENS were brought on board. Moreover, the selection should not be exposed to prevailing disasters. The total lengths of viable redundancy routes were then estimated. Because transmission line construction costs are quoted per km length and based on voltage levels, redundancy options with longer line lengths and higher voltage ratings will cost more than those with shorter lengths and lower voltage levels. However, the cost-benefit analysis of redundancy options was not part of this analysis due to data unavailability challenges. Fig. 2 shows a summary of the simulation process to achieve transmission line redundancy, where PF stands for PowerFactory environment.

3. Results and discussions

Implementing the transmission line redundancy plan has yielded significant improvements in grid resilience, as evidenced by the analysis in this section. This section presents the key findings from our comprehensive assessment, highlighting the impact of redundancy measures on the transmission lines that contribute to energy not being supplied (TLENS). Through TLENS metric and comparative analysis, we demonstrate the effectiveness of the strategies implemented to improve the electrical grid's reliability and robustness. The results underscore the critical role of redundancy in maintaining a stable and resilient power supply.

3.1. Transmission line criticality assessment to determine redundancy needs (redundant line location)

Results suggested that some lines (Nanjoka – Nkhotakota, Dwangwa – Chintheche, Nkhotakota -Dwangwa and Kapichila – Mlambe) contributed more to ENS per annum (shown by the red map on a geographic diagram in Fig. 3) than others, presenting a resilience risk.

Panteli also reported the identification of critical lines (Panteli et al., 2015a) where the corridor connecting the North GB to South GB was identified as a crucial line and some lines in the South GB for supplying London. However, there was no mention of how this was identified. Although (Yang and Li, 2022) described using a corridor-level resilience index to identify critical lines, the specific metric was not mentioned. In addition, a system-level resilience indicator was compared with a standard value to determine the vulnerability of transmission corridors (Zhao et al., 2021). used the probability of failure of each transmission line to identify weak links under ice disasters. This does not quantify how the failure of the weak links affects electricity delivery. In this case study, the inability of Nanjoka – Nkhotakota, Dwangwa – Chintheche and Nkhotakota – Dwangwa would affect the whole Northern region, affecting approximately 17 per cent of the customers connected to the grid. This was validated by the physical damage report (ESCOM, 2022b) following the 2022 Tropical Cyclone Ana, confirming the need for redundancy of these lines or corridors. During the 2022 TCA, the 132 kV Kapichila – Mlambe, 132 kV Golomoti – Salima, and 66 kV Mapanga – Fundis, Liwonde – Ntaja, Nkula – Blantyre and Nkula – Mapanga, causing power losses in Nchalo, Bangula, Nsanje, Mulanje, Phalombe, part of Chiradzulu, Ntaja, Nsanama, Nkaya, some parts of Blantyre, Zomba and Salima going to the North. Fig. 4 depicts the impact of TCA on major load centres following the failure of both generation and

Table 4
Possible redundant routes for critical lines based on existing substation voltage levels and road networks.

Critical line	Destination node	Routes depending on existing road network		Type redundancy	Enhancement scenario (ES)
		i-terminal	j-terminal		
Nanjoka-Chintheche	Chintheche	Kanengo	Chintheche	Loop	ES1
		Nanjoka	Chintheche	Parallel	ES2
		Nkhoma	Chintheche	Loop	ES3
	New Bwengu	Tsabango	New Bwengu	Loop	ES4
	Nkhotakota	Tsabango	Nkhotakota	Loop	ES5
Mapanga-Fundis	Fundis	Chigumula	Fundis	Loop	ES6
		Mapanga	Fundis	Parallel	ES7
Kapichila-Mlambe	Mlambe	BT West	Mlambe	Loop	ES8
		Kapichila	Mlambe	Parallel	ES9
All critical lines	Various load centres	Various terminals		-	ES10

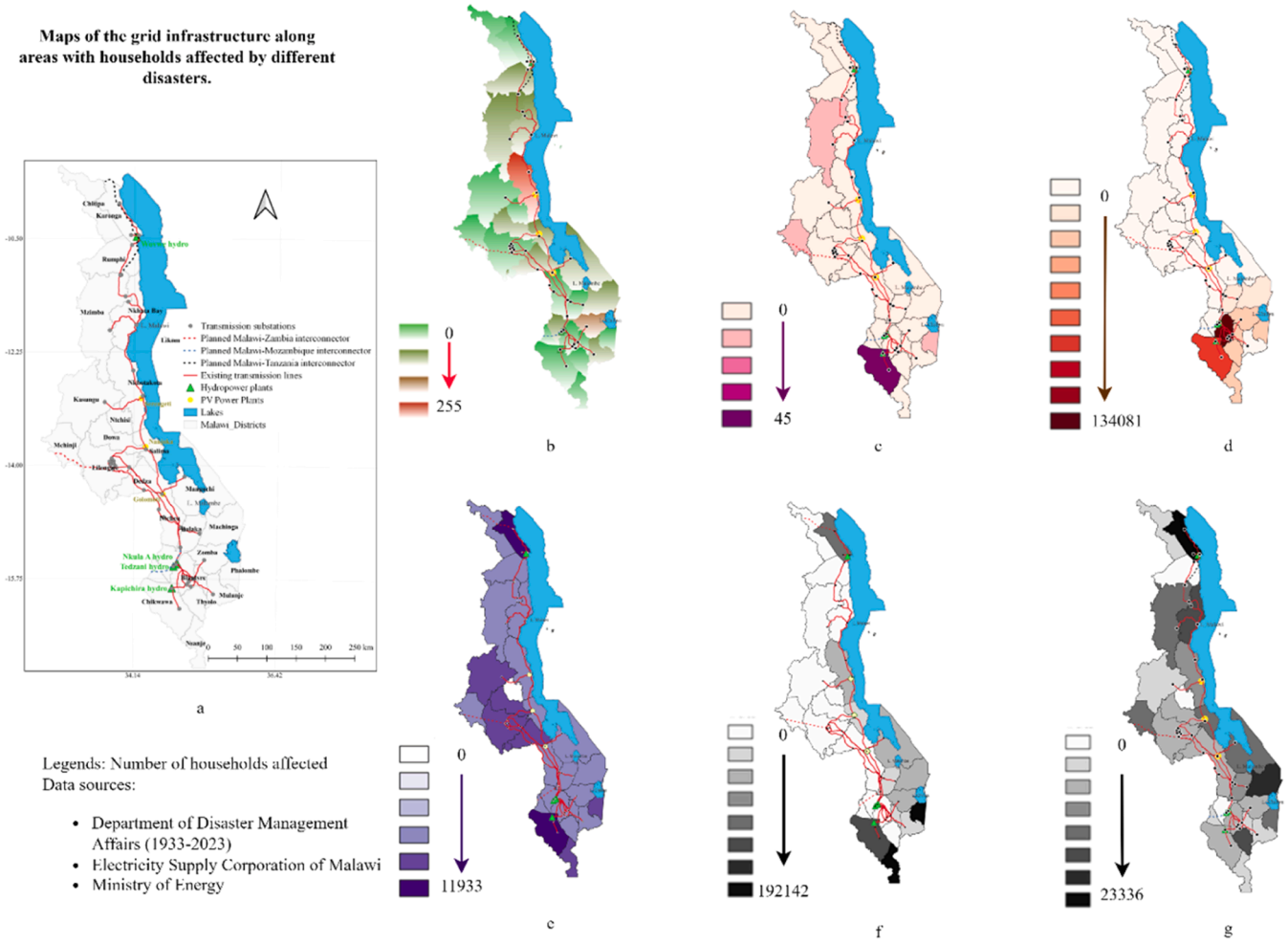


Fig. 5. Grid exposure to multiple disasters. (a) Map of Malawi showing grid infrastructure, (b) Fire, (c) Lightning, (d) Cyclone, (e) Drought, (f) Flood and (g) Wind.

transmission infrastructures. Although the Northern backbone (Nanjoka – Chintheche) was not physically affected by the 2022 TCA, the whole North had no electricity due to reduced generation capacity and the unavailability of redundant lines. It was easy to disconnect the entire North from Nanjoka. Failure of critical lines was implied to have resulted in some buses having increased unserved loads. The nodes with the highest unserved load required redundant generation supply (Kushal and Illindala, 2020). Failure of critical transmission lines during the 2022 TCA resulted in delayed restoration of electricity delivery. Although about 80 per cent of the generation was restored within 24 hours following damages caused by the Cyclone, it took up to 25 days to fully restore the network, leading to unserved loads in some major load centres.

3.2. Redundancy planning

Having identified the critical lines, potential redundancy routes were developed based on network topology, access to existing road infrastructure, TLENS, disaster maps and stakeholder consultations. The use of multiple strategies in formulating redundancy strategies is consistent with (Yang and Li, 2022), where reinforcement costs (based on the line lengths and strengthening costs), construction difficulty and the improvement effect were regarded as factors to be considered when formulating redundancy strategies, although the construction difficulty

Table 5
Summary of districts exposed to multiple disasters.

District	Disasters					
	Windstorms	Cyclones	Floods	Drought	Lightning	Fire
Balaka						
Blantyre		✓				
Chikwawa		✓	✓	✓	✓	
Chiradzulu						
Chitipa						
Dedza						
Dowa						
Karonga			✓	✓		
Kasungu						
Likoma						
Lilongwe	✓					
Machinga						
Mangochi						
Mchinji						
Mulanje	✓					
Mwanza						
Mzimba					✓	
Neno						
Nkhata Bay						
Nkhotakota						✓
Nsanje		✓	✓			
Ntcheu						
Ntchisi						
Phalombe			✓			
Rumphi						✓
Salima	✓					
Thyolo						
Zomba						✓

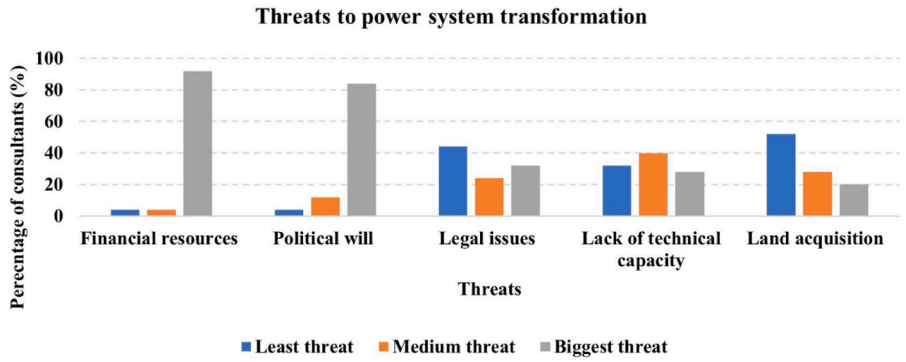


Fig. 6. Challenges to Structural Resilience Source (Chivunga et al#, 2024).

was not demonstrated.

3.2.1. Redundancy routes

For every identified critical line and their respective destination buses, possible routes were proposed by following existing road networks as indicated in Table 4. Critical lines, destination nodes, i-terminal, j-terminal, type of redundancy and enhancement scenario (ES) are presented.

3.2.2. Based on exposure (disaster maps)

Malawi's electricity grid is exposed to multiple disasters, including drought, floods, lightning strikes, cyclones, fires, and wind. Mapping was done using a quantum geographic information system (QGIS). Malawi disaster profile (1933 – 2023) and power system data were used. Disaster data was collected from the Department of Disaster Management Affairs. Geospatial data for transmission lines, transmission substations, and other generating plants were collected from the grid operator. The planned transmission system was acquired from the Ministry of Energy. Geospatial data for the electricity generating stations were mapped using Google Maps. Some districts are prone to disasters, especially in Chikwawa, Nsanje, Phalombe, and Karonga districts, as shown in Fig. 5.

Of these, cyclone disasters, associated with high rainfall intensity and wind loadings, were identified as extreme events following substantial physical damage and economic impacts on the generation and transmission power system. While efforts are being made to improve the resilience of the transmission system from tropical cyclones by building redundancy, there should be deliberate measures to not subject the redundancy options to other existing disasters, which can affect the transmission system. The parallel redundancy option to Mlambe (Chikwawa district) was not assessed because it was assumed to be exposed to the same conditions. The better option is either making the existing line more robust (Panteli et al., 2015a) or having distributed generation. BT West-Mlambe (ES6) is a better option for providing an alternative supply source to Kapichila without distributed generation. From the Table 5, the risk-free redundancy line will be the eastern backbone, along the M1 (passing through Kasungu, Mzimba and Mzuzu).

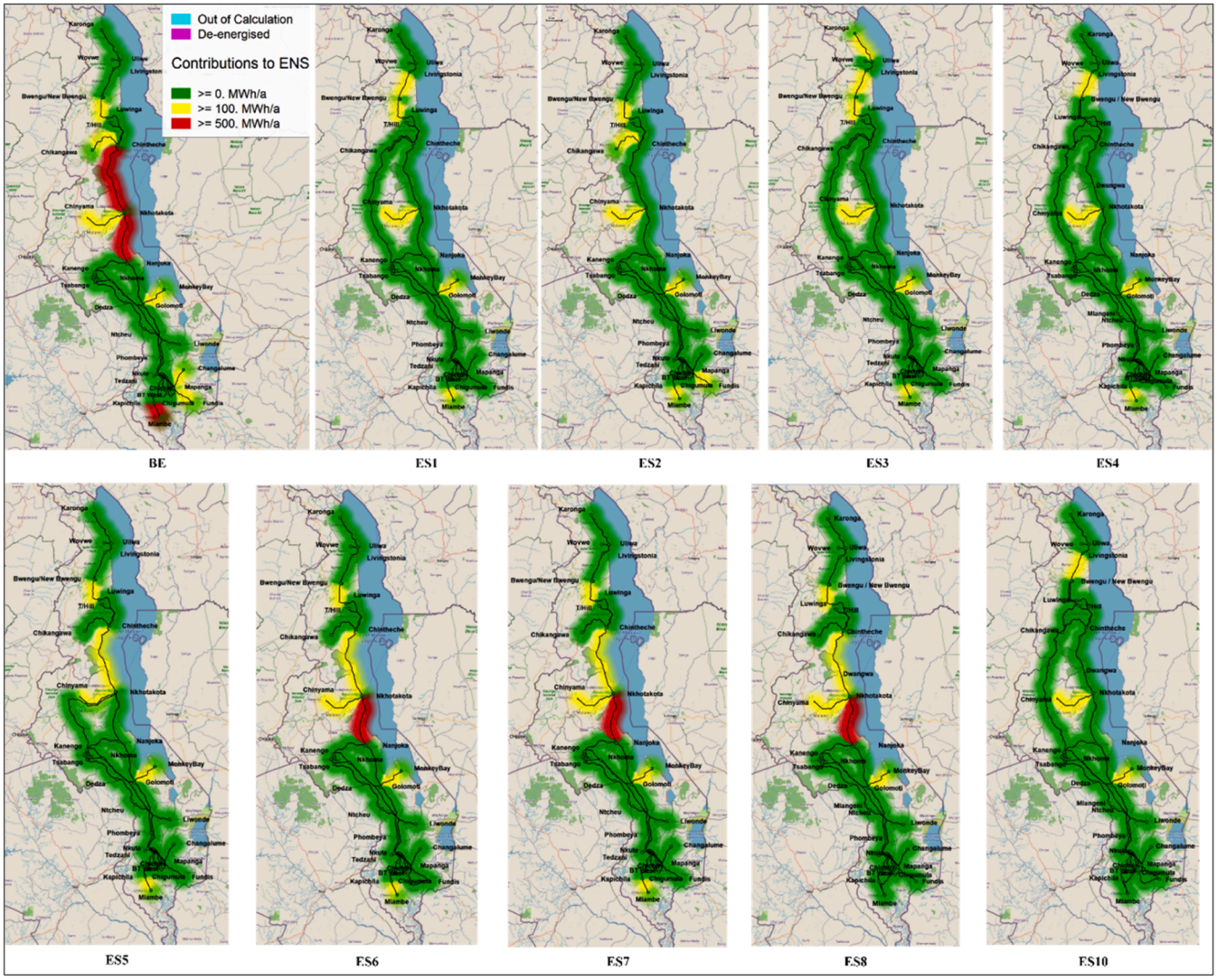


Fig. 7. Transmission lines contributing to ENS in MWh per annum for different scenarios. **BE**- before enhancement, **ES1** – Kanengo-Chintheche, **ES2**- Nanjoka-Chintheche redundant, **ES3**-Nkhoma-Chintheche, **ES4**-Tsabango-New Bwengu, **ES5**-Tsabango-Nkhotakota, **ES6**-Chigumula-Fundis, **ES7**-Mapanga-Fundis, **ES8**-BT West-Mlambe and **ES10**-Combination of ES4, ES6 and ES8. **Key: BE** – before enhancement and **ES** – enhancement scenario.

3.2.3. *Considering stakeholder consultations*

The lack of financial capacity was deemed the major challenge, as shown in Fig. 6, motivating our decision to model redundancy based on existing substations and road networks (Chivunga et al., 2024). The Operator is a state-owned entity which operates as a commercial entity to support its operations and capital budgets. However, resilience investments were too expensive to be supported by operational revenue (Chivunga et al., 2024). Recovery time and general resilience were associated with investment in compliance and enforcement programs (Shen et al., 2021). According to the qualitative study, Politics was believed to have contributed to the current derailed development in the power sector. Structural resilience measures are capital-intensive, and they take time to implement. Five (5) year political cycles do not match the energy investment cycle. This needs measures that would be implemented within a political cycle. However, the outcomes of the study conducted in (Chivunga et al., 2024) guided the choice of redundancy scenarios. Firstly, we planned around existing substations and road networks. Secondly, the line was prioritised to the North (Nanjoka-Chintheche) because the North grid was easily disconnected from the Centre (Nanjoka substation). Thirdly, we opted for an alternative route to the North (ES4) instead of the parallel redundancy (ES3), although both had the lowest TLNES due to the wind-related outages along the Nanjoka-Chintheche route. Finally, selecting BT West-Mlambe scenario (ES8) and Chigumula-Fundis (ES6) even though some portions will still be exposed to the same weather conditions because the Mlambe and Fundis substations supply high economic activities and there is no distributed generation.

3.3. *Effectiveness of redundancy measures*

The lowest ENS contribution was observed using a combination of Tsabango-New Bwengu (ES4), Chigumula-Fundis (ES6) and BT West-Mlambe (ES8) lines, as shown in Fig. 7 and Table 6. Reduction in resilience indicators, including loss of load expectation, loss of load frequency and unserved load, was observed across all contingencies following network redundancies (Panteli et al., 2015a; Kushal and Illindala, 2020). However, although Mapanga-Fundis and BT West-Mlambe reduced the ENS contribution for Mapanga-Fundis and Kapichila-Mlambe critical lines, these areas are exposed to cyclone disasters. Thus, using an alternative route to Fundis from Chigumula may be a less risky approach. Redundant lines are also similarly vulnerable to extreme weather conditions (Panteli et al., 2015a). Moreover, since the whole of Lower Shire (Chikwawa and Nsanje districts) is a disaster-prone zone, having an alternative supply to Mlambe substation

Table 6
TLENS for different lines under different enhancement scenarios.

Line name.	Contribution to ENS [MWh/a]									
	Before enhancement (**BE**)	After enhancement								
		Kanengo-Chintheche (**ES1**)	Nanjoka-Chintheche redundant (**ES2**)	Nkhoma-Chintheche (**ES3**)	Tsabango-New Bwengu (**ES4**)	Tsabango-NKK (**ES5**)	Chigumula-Fundis (**ES6**)	Mapanga-Fundis (**ES7**)	BT West-Mlambe (**ES8**)	Combination of ES4, ES6 and ES8 (**ES10**)
Nanjoka - Nkhotakota	2139.98	28.91	5.62	6.15	5.32	5.96	841.47	846.50	846.43	4.65
Dwangwa-Chintheche	1265.52	18.17	22.54	5.14	4.29	488.15	472.56	475.38	475.23	3.64
Nkhotakota-Dwangwa	751.35	21.94	33.65	5.75	4.99	340.48	329.26	331.23	331.16	4.792
Kapichira - Mlambe	516.11	173.30	200.87	168.13	173.36	174.10	173.67	174.74	3.08	0.841
Serengeti-Chinyama	385.40	122.65	377.99	141.20	121.59	116.22	113.69	114.34	114.16	120.124
Mapanga - Fundis Cross	346.96	78.55	217.15	77.72	78.38	79.49	4.03	3.56	80.32	1.41
Golomoti - MonkeyBay	213.30	169.79	158.68	198.85	172.00	171.15	169.49	170.49	170.41	170.67
Luwinga-Bwengu	202.95	197.87	141.35	229.89	2.94	166.46	161.63	162.58	162.48	2.58
Mapanga - Changalume	192.43	80.14	76.33	79.11	80.04	80.98	81.59	82.03	81.71	79.77
Nkhoma-Tsabango	184.18	6.46	7.06	7.96	1.11	1.89	7.62	7.65	7.56	0.87
Chinyama -Shayona	131.96	42.01	129.42	48.34	41.63	39.79	38.93	39.15	39.09	41.13
Chintheche - Chikangawa	126.66	2.71	260.54	2.82	2.05	4.14	5.74	5.74	5.53	1.498
Livingstonia - Bwengu	123.39	115.82	81.54	134.53	103.78	98.43	96.27	96.83	96.69	102.33
Uliwa - Karonga	114.18	88.54	60.15	102.47	78.99	75.75	74.73	75.15	74.97	77.88
Line lengths (km)	0	323	283.32	348	421.75	237	54.71	44.012	64.123	540.582

and the whole Lower Shire offers more resilient benefits than the redundant line. ES1 and ES3 provide loop connections to Chintheche, while ES2 provide a parallel redundant connection. ES4 provides both loop connection and an alternative route to the North through New Bwengu substation from Tsabango. Table 6 shows the different levels of effectiveness of these redundant scenarios. Taking the Nanjoka-Nkhotakota line for example, the alternative route (ES4) gives the lowest TLENS compared with loops and parallel redundant scenarios. Although the parallel redundant scenario (ES2) is lower than the loops (ES1 and ES3), this ES2 is exposed to fire and wind hazards, making it a less favourable scenario.

The exact costs of transmission line redundancy and cost-benefit analysis of redundancy scenarios are outside the scope of this work. However, suppose assumptions are made as in (Yang and Li, 2022), where the transmission line cost was fixed at 1 million USD/km. In that case, it is estimated that ES10 will be more costly than the individual scenarios, emphasising the cost implications of resilience improvement. Nonetheless, the benefits of resilience investments cannot be overemphasised. For a comprehensive cost-benefit analysis of the redundancy scenarios, we suggest that the preliminary steps would be to have adequate and accurate cost data because most studies use assumed values, which might not be a true reflection of the actual costs due to differences in terrain, construction difficulties and specific resilience requirements. Since Malawi's power system is not highly diverse and the solar PV plants do not take part in frequency response, high redundancy and low response diversity can effectively coexist (Xu et al., 2023). Thus, ES10 is selected because it offers the lowest TLENS. This aligns with (Yang and Li, 2022), where the best redundancy strategy was based on the best resilience improvement effect. Fig. 8 summarises the effectiveness of using enhancement scenario 10 (ES10). Visualisation of the power system model after enhancement is presented in Fig. 9. This model only presents those lines at 132 kV for simplicity and readability. The selected enhancement scenario provides a ring network (alternative route) to the North grid, Lower Shire and Fundis-Cross. The new redundancy line to the North grid is not exposed to fire and wind disasters, although a small portion is vulnerable to lightning disasters. However, the grid infrastructure is protected against lightning strikes (Chivunga et al., 2024). Although BT West to Mlambe (ES6) provides an alternative route to the Lower Shire, the topography of the area requires that this redundant line passes through the same location, which is exposed to floods, cyclones and lightning strikes, underscoring the need for disaster-proof transmission lines. The interest was to change the red lines. Although other lines were also in yellow, based on engineering judgement and stakeholder consultations, redundant lines, such as the Mapanga-Fundis, were suggested because of huge economic activities in tea estates. Some lines, such as Golomoti – Monkey Bay, Serengeti – Chinyama and New Bwengu – Livingstonia, are still in yellow. Running redundant lines for these may not offer an economic advantage, considering that there are few heavy economic activities in the load centres these lines supply. Although operational resilience enhancement measures are less expensive than structural measures, such as redundancy and robustness, these infrastructural measures make the grid more robust and bigger (Panteli and Mancarella, 2015a). Of course, the typical lengths of overhead lines will significantly add to the costs. Hence, such approaches can be implemented for sections of the circuit that are more at risk. Another study identified financial constraints as the biggest threat to resilience enhancement (Chivunga et al., 2024). As such, redundancy must be afforded during times without severe disruptions (Soikkeli et al., 2023). Network redundancy introduces flexibility and robustness. It also increases the success rate of corrective measures (Daeli and Mohagheghi, 2022). However, distributed energy resources may be an option for implementing the Malawi 2063 agenda.

The effectiveness of transmission line redundancy.

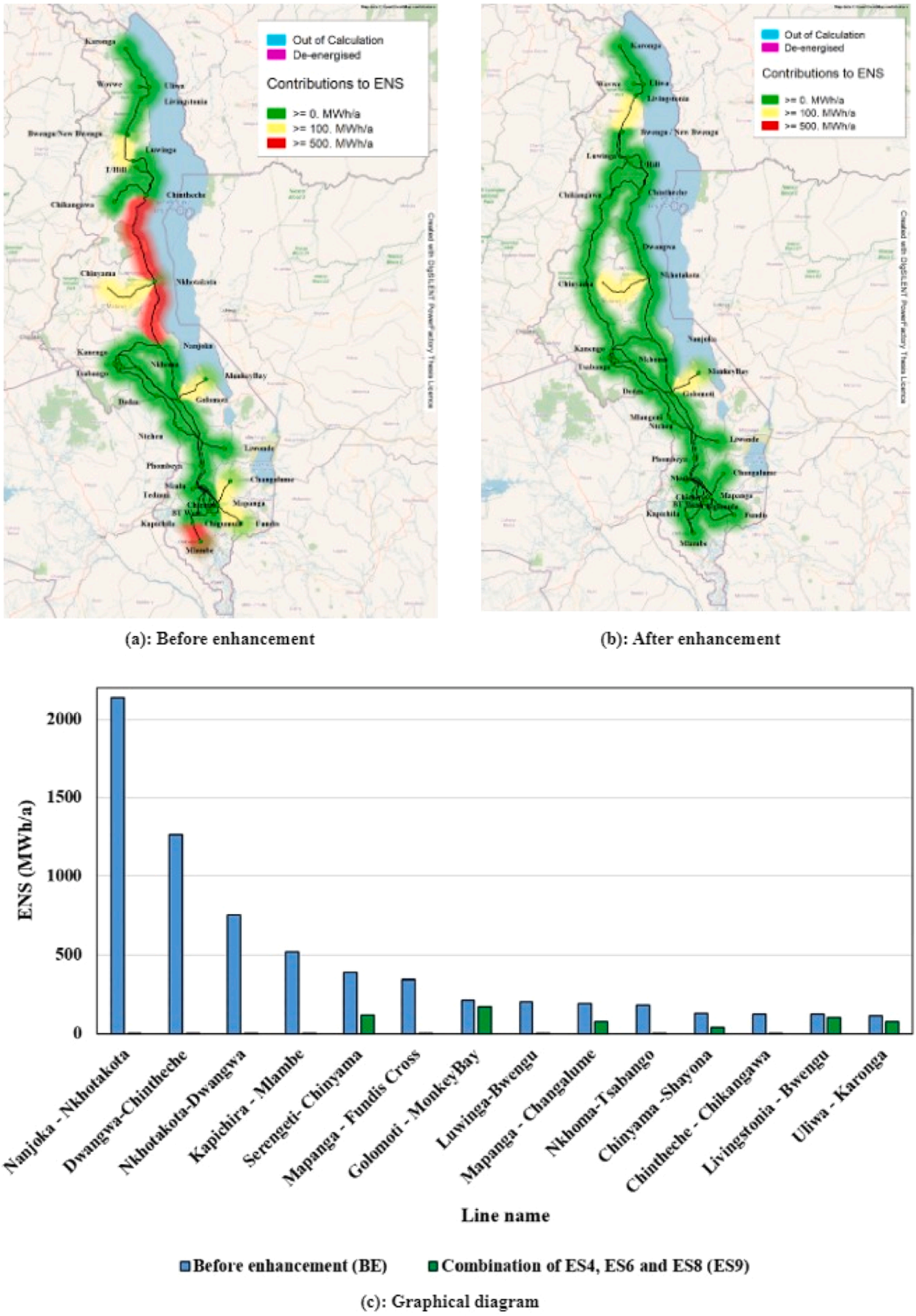


Fig. 8. TLENS before and after redundancy.

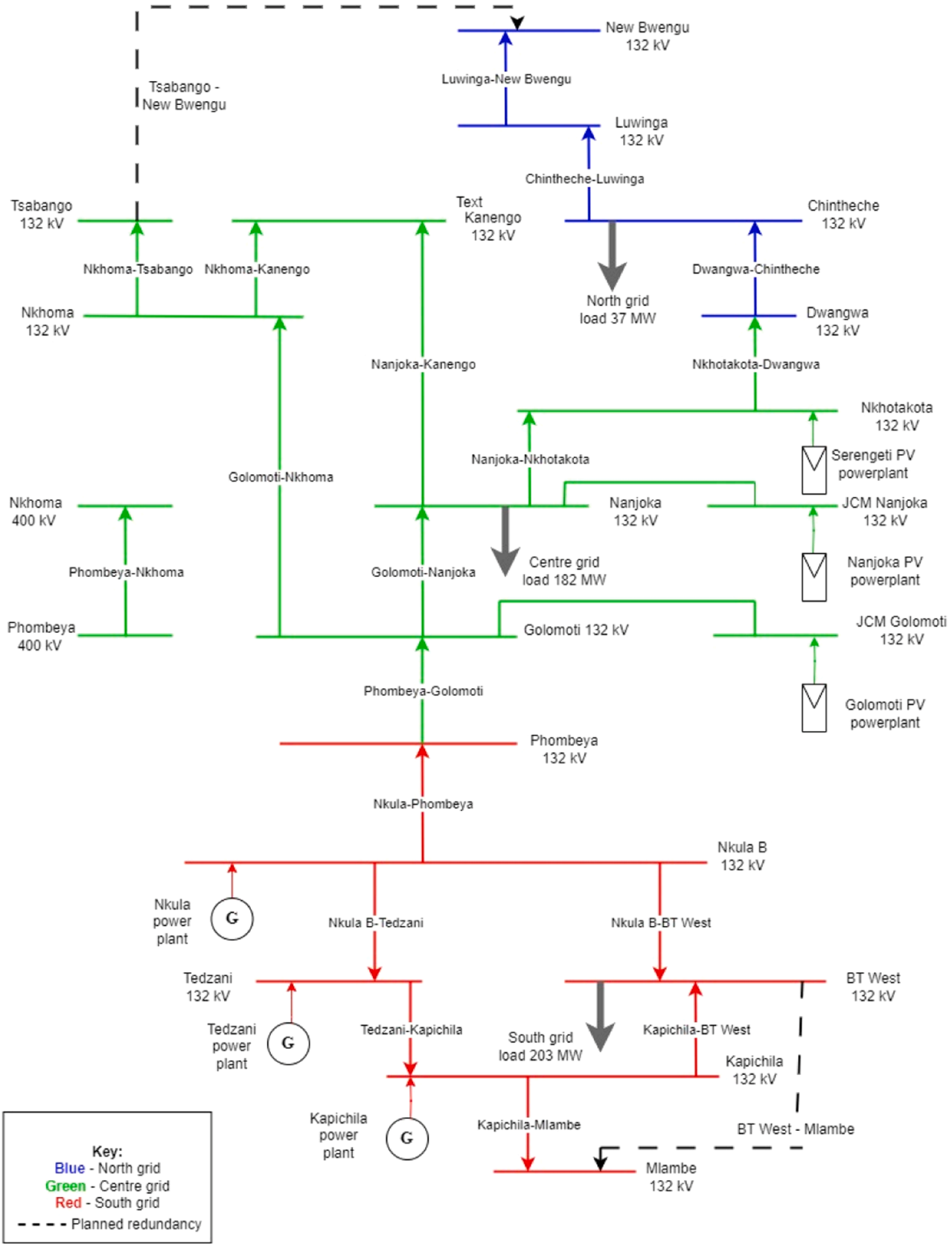


Fig. 9. 132 kV transmission power system model after redundancy.

4. Conclusion

This study employed the Transmission Lines contributing to Energy Not Supplied (TLENS) indicator as a quantitative guide to improve network resilience via directed transmission line redundancy. The primary purposes were to assess the criticalness of transmission lines, develop and evaluate redundancy strategies, and use these strategies in a realistic case study. First, the examination of the existing transmission network exposed that the lines linking the central grid to the north grid and the line to the Lower Shire are the utmost critical, unveiling the peak TLENS values and playing a fundamental role in grid resilience and reliability. Next, the redundancy planning was carefully designed and centred on criticality evaluation, historical exposure to catastrophes, and ease of access via existing road networks. This broad methodology guaranteed that redundancy techniques were effective but also realistic and feasible. The outcomes reveal a significant effect of these redundancy measures on lessening the contributions to ENS. Although parallel routes were established to lessen TLENS contributions more than loop networks, the latter was eventually preferred due to their reduced exposure to catastrophes, as demonstrated by our disaster mapping analysis. This preference emphasises the value of reflecting on environmental and logistical influences alongside entirely quantitative approaches in redundancy planning. A trade-off has to be made between the cost of redundancy and its resilience benefit. Some areas with minimal economic activities may benefit from distributed generation (local redundancy). In summary, using the TLENS metric in evaluating and improving network resilience has shown to be effective in our case study, presenting significant perceptions for potential network resilience enhancements. The strategic application of redundancy measures designed for critical lines with high TLENS values can significantly reduce the impact of line failures, thus warranting a more stable and reliable energy supply. This approach and its outcomes provide a robust framework for ongoing attempts at enhancing grid resilience. In future works, generation supply redundancy will be considered, highlighting the need for viable locations.

CRediT authorship contribution statement

Joyce Nyuma Chivunga: Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Fransisco Gonzalez Longatt:** Writing – review & editing, Validation, Software. **Zhengyu Lin:** Writing – review & editing, Supervision. **Richard Blanchard:** Writing – review & editing, Visualization, Supervision, Methodology.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The data that has been used is confidential.

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